

Step-by-Step Solutions: Statistics Problems

Problem 7: Unemployment Benefits

Data (15 values):

239, 214, 327, 416, 321, 289, 209, 356, 190, 272, 252, 272, 310, 276, 251

Range

$$\text{Range} = 416 - 190 = \mathbf{226}$$

Mean

$$\sum x_i = 4194, \quad \bar{x} = \frac{4194}{15} = 279.6$$

Variance and Standard Deviation

Compute squared deviations:

$$\sum (x_i - \bar{x})^2 = 50387.6$$

Sample variance:

$$s^2 = \frac{50387.6}{14} = 3577.6857$$

Sample standard deviation:

$$s = \sqrt{3577.6857} \approx 59.81$$

Final Answers

$$\text{Range} = 226, \quad s^2 \approx 3577.69, \quad s \approx 59.81$$

Problem 19: Cost per Load of Laundry Detergent (Grouped Data)

Class Midpoints

16, 23, 30, 37, 44, 51, 58, 65

Mean

$$\sum fm = 1183, \quad N = 35$$
$$\bar{x} = \frac{1183}{35} = 33.8 \text{ cents}$$

Variance

Compute:

$$\sum f(m - \bar{x})^2 = 4541.6$$

Population variance:

$$\sigma^2 = \frac{4541.6}{35} = 129.76$$

Population standard deviation:

$$\sigma = \sqrt{129.76} \approx 11.39$$

Final Answers

$$\text{Mean} = 33.8, \quad \sigma^2 = 129.76, \quad \sigma = 11.39$$

Problem 34: Calories (Chebyshev)

Mean: $\mu = 240$, SD: $\sigma = 38$.

We want at least 75%. Solve:

$$1 - \frac{1}{k^2} \geq 0.75 \Rightarrow k = 2$$

Range:

$$240 \pm 2(38) = [164, 316]$$

Final Answer

At least 75% of the data lie in:

$$[164, 316]$$

Problem 35: Time Spent Online (Chebyshev)

Mean: $\mu = 180$ minutes, SD: $\sigma = 32$.

We want at least 88.89%.

$$1 - \frac{1}{k^2} \geq 0.8889 \Rightarrow k = 3$$

Range:

$$180 \pm 3(32) = [84, 276]$$

Final Answer

At least 88.89% lie in:

[84, 276] minutes

Problem 38: Maze Trials (Chebyshev)

Mean: $\mu = 12$, SD: $\sigma = 3$. Given interval: 4 to 20.

Distance from the mean:

$$k = \frac{8}{3}$$

Chebyshev bound:

$$1 - \frac{1}{k^2} = 1 - \frac{9}{64} = \frac{55}{64} = 0.859375$$

Final Answer

At least:

$$85.9375\% \approx 85.94\%$$

of values lie in the range [4, 20].